Timothy Mitchell

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EDUCATION

University of North Carolina at Charlotte, Charlotte, NC
Bachelor of Science in Mechanical Engineering, Concentration in Motorsports Expected Graduation May 2028

Durham Tech Community College, Durham, NC
Associate of Science in Engineering May 2025

RELATED COURSEWORK:

Statics | Solids | Mechatronics 1 | Manufacturing Systems | Machine Shop | Dynamics | Thermodynamics | Mechatronics 2 | Linear Algebra | Fluid Mechanics | Mechanics of Materials | Computational Methods

SKILLS:

SolidWorks, Onshape, **Python**, MATLAB, Java, React, C, C++, Visual Studio Code, Excel, Microsoft Tools, Computer Hardware, 3D modeling, 3D design, Manufacturing, **3D printing**, Arduino, ESP32, KiCAD, ANSYS, **Mastercam**, Vericut, **MoTeC**, Manual Mill and Lathe, CAD, **FEA**

ENGINEERING EXPERIENCE

Formula Student Vehicle Dynamics Design Team Member, 49ers Racing, Charlotte, NC January 2026 – Present

- Designed/Validated Suspension Components in SolidWorks/Ansys, raising the safety factor(SF) and reducing weight by 27%
- Used Excel to build an Anti-Roll Bar (ARB) calculator to determine a roll stiffness variability of +/- 20%
- Used MoTeC to build worksheets tracking 20+ performance targets and drive data-based setup changes.
- Built a 400+ part SolidWorks assembly of the 2027 car to validate suspension components and hardware.
- Performed an Ansys Root Cause Analysis (RCA) on a Shock bar, finding an SF of 1, then redesigning the components, doubling the SF

Manufacturing Engineering Intern, Northeast Tool and Manufacturing, Matthews, NC August 2025 – Present

- Designed, validated, and manufactured a transformer tower in 5 days to deliver on a critical deadline.
- Worked in SolidWorks to develop complex fixtures for 5-axis CNC machines.
- Produced critical fixtures >.001” using Mastercam to develop 5-axis toolpaths for DMG MORI machines.
- In >2 months, participated in Kaizen events to optimize and modernize our shipping department from scratch.
- Developed custom software tools integrating with our ERP's API to improve efficiency and organization by 12%.

Lead Mentor, Codewiz FTC Team, Durham, NC August 2023 – August 2025

- Organized grants and secured over $2,000 in funding to cover 2 years of team expenses.
- Led a team of 10 students to build a robot in 4 months and compete in several FTC competitions.
- Taught advanced CAD principles and Java programming to students in 4 months.

PROJECTS

Junior Design Project, Robotic Design Competition, Charlotte, NC January 2026 – May 2026

- Designed the competition's first> 1 lb robot using SolidWorks topology optimization to design and validate
- Won the categories for the lightest by over a lb, smallest, and cheapest robot
- Designed the electronics and programmed the ESP32 microcontroller working 100% of the time.
- Used 3D printing creatively to produce topology-optimized parts, reducing the weight by 30% and functional GT3 belts out of TPU.
- Worked in a team of 7 to design, validate, manufacture, and test a robot for the competition challenge.

Technical Lead, Rival Robotics Competition Team, Raleigh, NC August 2024 – Present

- Designed a 3D model in Onshape of a competition robot based on defined criteria and constraints.
- Incorporated manufacturability and machinability into the robot design using bent sheet metal and waterjetting
- Developed in < 1month a website using React to showcase the team's accomplishments and information.

OTHER EXPERIENCE

Center Director / Manager August 2023 – August 2025
Code Wiz Durham, Durham, North Carolina

- Led and trained a team of 10+ coaches, managing scheduling and performance for team success.
- Organized and ran school partnerships with 90+ students, teaching robotics in classroom settings of 15+ students.